TX-OMICS REVISION

Seven gene sets.
Seven clear definitions.

Family 1 captures four primary evidence categories. Family 2 contains three CSW-derived GO term groups, with FC0.5 and FC1 provenance shown separately.

4
primary sets

3
GO-derived sets

2
thresholds

7
stocker CSVs

Two families, seven non-interchangeable sets

Each set now receives one slide; overlap is reported separately within each family.

FAMILY 1 · PRIMARY EVIDENCE

01	6	**Mechanistic**	Consistent/correlated + published homeostatic-consistent effect
02	20	**Homeostatic genes**	Opposing history versus rebound correlations
03	97	**CSW 4+ genes**	Consistent in four or more of seven datasets
04	4	**HLH genes**	Potential upstream regulators

FAMILY 2 · CSW-DERIVED GO

05	99	**Ribosomal**	FC0.5 99 · FC1 12
06	167	**Mitochondrial**	FC0.5 167 · FC1 23
07	5	**Immune**	FC0.5 0 · FC1 5

FC0.5 and FC1 counts are source-provenance counts. A gene may appear at both thresholds.

Mechanistic · 6 genes

Consistent or correlated sleep/wake genes with published effects compatible with homeostatic feedback.

6

paper-highlighted genes

Inclusion requires transcriptomic evidence plus published sleep/wake function.

- Consistent across sleep/wake conditions or correlated with sleep/wake history
- Published perturbation changes sleep/wake
- Effect direction fits a potential homeostatic loop

unc79
NALCN complex

SIFa
neuropeptide

rumpel
glial transporter

AstA-R2
neuropeptide receptor

Trhn
serotonin synthesis

RpL23
ribosome / translation

These six are not the 20 opposing history-rebound correlation genes.

Homeostatic genes · 20 genes

Genes correlated with pre-sampling sleep/wake history and post-sampling rebound in the opposite direction.

15

History+ / Rebound−

Positive correlation with prior sleep; negative correlation with rebound.

5

History− / Rebound+

Negative correlation with prior sleep; positive correlation with rebound.

15 · HISTORY+ / REBOUND−

AstA-R2 · bumpel · CG33080 · CG41378 · CG7460
CG7601 · Chrac-14 · Cyp28c1 · dpr21 · kumpel
Obp44a · Sk1 · Ssk · Trhn · VGlut2

5 · HISTORY− / REBOUND+

CG42323 · CG7079 · CG9313 · CG9377 · Gtpbp1

CSW 4+ genes · 97 genes

Consistent sleep/wake genes evident across four or more of the seven datasets.

97

FC0.5 wake genes

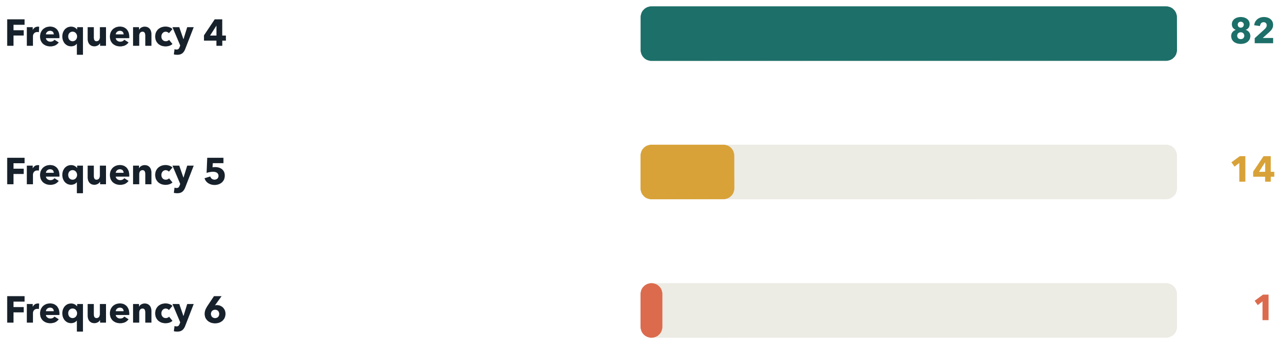
The category rule is recurrence across datasets—not GO enrichment.

7

also qualify at FC1

All seven are already contained within the 97-gene FC0.5 set.

Recurrence across seven datasets



No sleep-upregulated gene met the ≥4-of-7 recurrence criterion.

HLH genes · 4 genes

Four bHLH factors identified as potential upstream regulators of wake-induced gene expression.

REGULATORY LOGIC

CAGCTG

E-box enriched upstream of wake-induced genes

HOMER motif enrichment → candidate direct regulators

Results · Table S1

bigmax

FBgn0039509

HLH3B

FBgn0011276

E(spl)m3-HLH

FBgn0002609

E(spl)mbeta-HLH

FBgn0002733

This set is regulatory evidence—not frequency or correlation evidence.

Ribosomal / translation · 99 genes

GO term–grouped hit genes with FC0.5 and FC1 source provenance shown separately.

THRESHOLD PROVENANCE

99

unique genes in this set

FC0.5	99	genes supported
FC1	12	genes supported
BOTH	12	appear at both thresholds

99 FC0.5 + 12 FC1 − 12 both = 99 unique

Most frequent approved terms

Ribosome	93
Structural constituent of ribosome	92
Translation	77
Cytoplasmic translation	65
Cytosolic ribosome	58

87 FC0.5-only · 0 FC1-only · 12 at both thresholds

Mitochondrial / metabolism · 167 genes

GO term–grouped hit genes with FC0.5 and FC1 source provenance shown separately.

THRESHOLD PROVENANCE

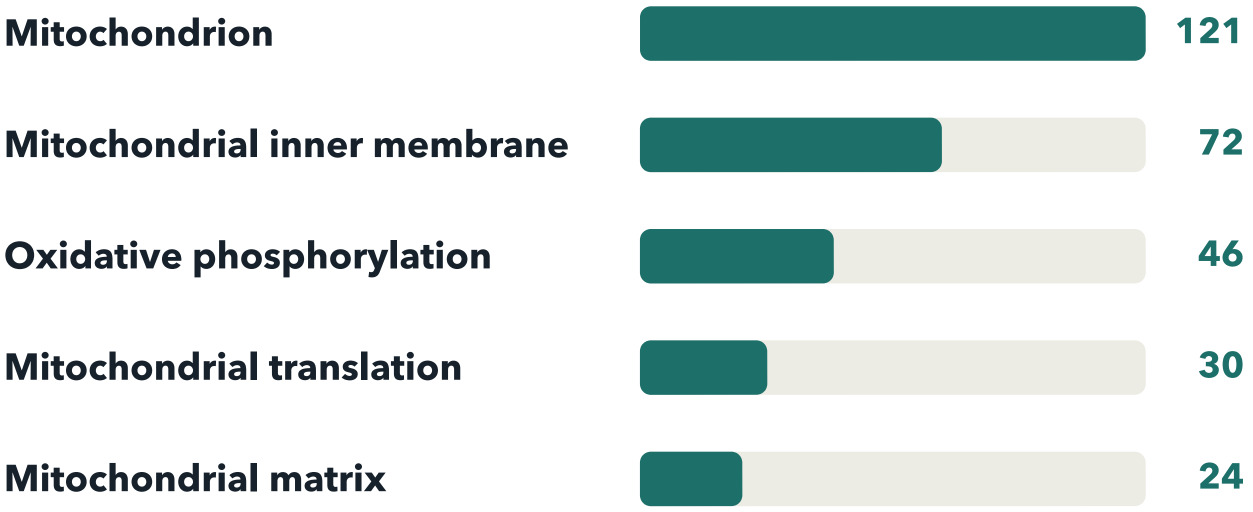
167

unique genes in this set

FC0.5	167	genes supported
FC1	23	genes supported
BOTH	23	appear at both thresholds

167 FC0.5 + 23 FC1 − 23 both = 167 unique

Most frequent approved terms



144 FC0.5-only · 0 FC1-only · 23 at both thresholds

Immune · 5 genes

The immune GO term group is FC1-exclusive in this analysis.

THRESHOLD PROVENANCE

5

unique genes in this set

FC0.5	0	genes supported
FC1	5	genes supported
BOTH	0	appear at both thresholds

0 FC0.5 + 5 FC1 − 0 both = 5 unique

Five FC1-supported genes

- BomBc2
- BomS1
- BomS2
- IM14
- Thor

Approved GO terms · genes per term

Defense response	4
Antibacterial humoral response	4
Response to bacterium	4

Genes may occur in more than one approved term.

0 FC0.5-only · 5 FC1-only · 0 at both

Only three genes cross primary-set boundaries

Overlap is calculated among Mechanistic, Homeostatic, CSW 4+, and HLH only.

Observed shared genes

Mechanistic × Homeostatic	2	**Trhn · AstA-R2**
Mechanistic × CSW 4+	1	**RpL23**
All other pairings	0	No shared FBgns

Family 1 union

124	unique FBgns
121	in exactly one set
3	in exactly two sets
0	in three or four sets

HLH genes are fully disjoint.

The GO-derived union contains 240 unique genes

99 ribosomal + 167 mitochondrial + 5 immune − 31 shared ribosomal/mitochondrial genes = 240 unique FBgns.

Union of Ribosomal, Mitochondrial & Immune · 240 unique FBgns

- **68** Ribosomal only
- **31** Ribosomal ∩ mitochondrial
- **136** Mitochondrial only
- **5** Immune only

Category overlap

31 shared genes retained in both Ribosomal and Mitochondrial

Threshold provenance across the 240-gene union

- **203** FC0.5 only
- **32** both thresholds
- **5** FC1 only

Only the five Immune genes are FC1-exclusive.